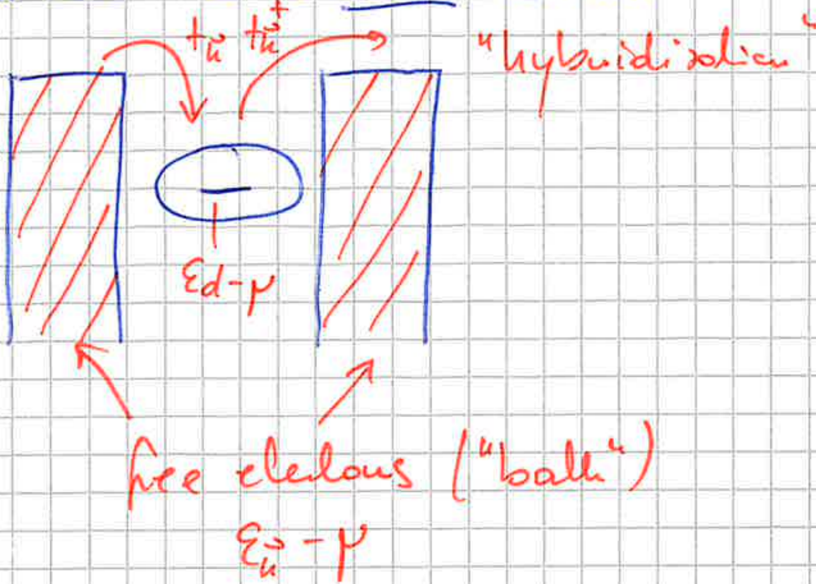
2.2 The Anderson impurity model

quantum dot with one atomic level ("d-electrons")



Taken together, we are left with the following Hamiltonian:

$$H = H_c + \underbrace{H_d + H_U}_{\text{"impurity"}} + H_{hyb}$$

Conduction electrons

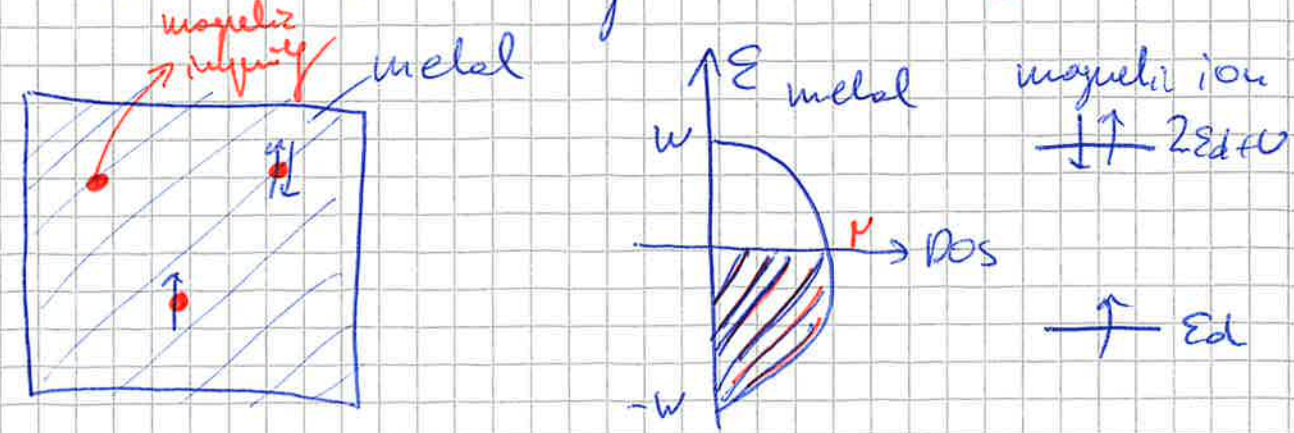
$$\sum_{\vec k\sigma}(\varepsilon_{\vec k}-\mu)c^\dagger_{\vec k\sigma}c_{\vec k\sigma}$$

$$\sum_{\sigma}(\varepsilon_d-\mu)c^\dagger_{d\sigma}c_{d\sigma} + U n_{d\uparrow}n_{d\downarrow}$$

hybridisation

$$\sum_{\vec k\sigma} t_{\vec k}\, c^\dagger_{d\sigma}c_{d\sigma} + \sum_{\vec k\sigma} t^*_{\vec k}\, c^\dagger_{\vec k\sigma}c_{\vec k\sigma}$$

This Hamiltonian is called the Anderson impurity model (AIM), because it was originally introduced to describe magnetic impurities embedded in a homogeneous host metal



Although the AIM looks very simple, its solution, in fact, is very complicated and has a rich phase diagram!

E.g.: for certain system parameters, the model has a magnetic moment (~~maximizing~~ (minimizing on-site interaction energy) or no magnetic moment (maximum in hybridization energy)

Apart from the rich phase diagram, the AIM is a useful tool for (i) analyzing the so-called Kondo problem and (ii) the Hubbard model (within the so-called dynamical mean-field theory).

The first physical question which we want to answer: when is the AIM magnetic?

For this analysis we will first introduce the equation of motion theory for the Green function.

2.2.1 Equation of motion (EOM) theory for the single-particle Green function

retarded fermionic Green function:

$$G^R(\vec{r}t,\vec{r}'t') = -i\Theta(t-t') < \{ \Psi(\vec{r},t), \Psi^+(\vec{r}'t') \} >$$

equation of motion: derivative w.r.t. first time argument

Let's take a general basis: $\{\nu\}$

$$H_0 = \sum_{\nu\nu'} t_{\nu\nu'} c_{\nu'}^{\dagger} c_{\nu}$$

$$G^R(\nu t, \nu' t') = -i\theta(t-t') \langle \{ c_\nu(t), c_{\nu'}^{\dagger}(t') \} \rangle$$

$$i\partial_t G^R(\nu t, \nu' t') = (-i)\left(i\partial_t \theta(t-t')\right) \langle \{ c_\nu(t), c_{\nu'}^{\dagger}(t') \} \rangle +$$

$$+ (-i)\theta(t-t') \langle \{ i\partial_t c_\nu(t), c_{\nu'}^{\dagger}(t') \} \rangle =$$

$$= \delta(t-t') \langle \underbrace{\{ c_\nu(t), c_{\nu'}^{\dagger}(t) \}}_{\delta_{\nu\nu'}} \rangle +$$

$$- i\theta(t-t') \langle \{ i\partial_t c_\nu(t), c_{\nu'}^{\dagger}(t') \} \rangle =$$

$$= \delta(t-t')\delta_{\nu\nu'} - i\theta(t-t') \langle \{ i\partial_t c_\nu(t), c_{\nu'}^{\dagger}(t') \} \rangle$$

What is the dependence in time of the annihilation operator?

We assume that H is time-independent $\partial_t H = 0$

→ Heisenberg equation of motion

$$i\partial_t \left[c_\nu(t) \right] = -\left[H, c_\nu \right](t) =$$

$$= -\left[H_0, c_\nu \right](t) - \left[V_{int}, c_\nu \right](t)$$

quadratic part (e.g. kinetic energy) — all interactions

$$\rightarrow -\left[H_0, c_\nu \right] = -\sum_{\nu'\nu''} t_{\nu'\nu''} \left[c_{\nu'}^{\dagger} c_{\nu''}, c_\nu \right]$$

Here, the following relation becomes useful:

$$[AB, C] = A[B, C] + [A, C]B =$$
$$= A\{B, C\} - \{A, C\}B$$

$$\Rightarrow [c^\dagger_{\nu'} c_{\nu''}, c_\nu] = c^\dagger_{\nu'} \underbrace{\{c_{\nu''}, c_\nu\}}_{\phi} - \underbrace{\{c^\dagger_{\nu'}, c_\nu\}}_{= \{c_\nu, c^\dagger_{\nu'}\} = \delta_{\nu'\nu}} c_{\nu''}$$

$$\Rightarrow -[H_0, c_\nu] = + \sum_{\nu'\nu''} t_{\nu'\nu''} \delta_{\nu\nu'} c_{\nu''} = \sum_{\nu''} t_{\nu\nu''} c_{\nu''}$$

$$\Rightarrow \boxed{\begin{array}{l} \sum_{\nu''} (i\delta_{\nu\nu''} \partial_t - t_{\nu\nu''}) G^R(\nu''t, \nu't') = \delta(t-t')\delta_{\nu\nu'} + D^R(\nu t, \nu't') \\ \text{with } D^R(\nu t, \nu't') = -i\theta(t-t') \langle \{-[V_{int}, c_\nu](t), c^\dagger_{\nu'}\} \rangle \end{array}}$$

corrections to free-particle Green function

(in fact, they are higher-order Green functions)

Hamiltonian not time-dependent $\Rightarrow$ only difference $t-t'$

$\Rightarrow$ Fourier transform

$$\sum_{\nu''} [\delta_{\nu\nu''}(\omega + i\eta) - t_{\nu\nu''}] G^R(\nu''\nu', \omega) = \delta_{\nu\nu'} + D^R(\nu, \nu'; \omega)$$

$$D^R(\nu\nu', \omega) = -i \int_{-\infty}^{\infty} dt\, e^{i(\omega + i\eta)(t-t')} \theta(t-t') \langle \{-[V_{int}, c_\nu](t), c^\dagger_{\nu'}(t')\} \rangle$$

Examples for the EOM technique:

•) non-interacting particles

$D^R = 0 \Rightarrow$ solve equation

$$\sum_{\nu''}\left(\delta_{\nu\nu''}(\omega+i\eta)-t_{\nu\nu''}\right)G_0^R(\nu''\nu',\omega)=\delta_{\nu\nu'}$$

inverse Green function: $(G_0^R)^{-1}(\nu\nu';\omega)=\delta_{\nu\nu'}(\omega+i\eta) - t_{\nu\nu'} =$

$$= (\underline{\underline{G_0}}^R)^{-1}_{\nu\nu'} \quad \text{matrix}$$

$\to (\underline{\underline{G_0}}^R)^{-1}\underline{\underline{G_0}}^R = \mathbb{1} \Rightarrow$ invert equation for finding Green function

diagonal basis: $t_{\nu\nu'} = \delta_{\nu\nu'}\varepsilon_\nu$

$$(\underline{\underline{G_0}}^R)_{\nu\nu'} = G_0^R(\nu,\omega)\delta_{\nu\nu'} = \frac{1}{\omega-\varepsilon_\nu+i\eta}\delta_{\nu\nu'}$$

•) single level coupled to continuum

$$H = H_0 + H_{hyb} + H_e \quad \Rightarrow \text{no interactions, } V_{int} = \emptyset$$

non-int. electron gas, hybridization, atomic level

$$H_0 = \sum_\nu \xi_\nu c_\nu^\dagger c_\nu \,,\; \xi_\nu = \varepsilon_\nu - \mu$$

$$H_e = \xi_0 c_e^\dagger c_e \,,\; H_{hyb} = \sum_\nu \big(\underbrace{t_\nu^*}_{=t_{\nu e}^*} c_\nu^\dagger c_e + \underbrace{t_\nu}_{=t_{\nu e}} c_e^\dagger c_\nu\big)$$

We are interested in finding the diagonal Greenfunction for the local state. For this we need two Greenfunctions:

$$G^R(l,l,t-t') = -i\Theta(t-t') < \{ c_l(t), c_l^+(t') \} >$$
$$G^R(v,l,t-t') = -i\Theta(t-t') < \{ c_v(t), c_l^+(t') \} >$$

for fermions (electrons).

We let v'' run over both the local state l and all continuum states (denoted by v):

$$\sum_{v''} \left[\delta_{vv''}(\omega + i\eta) - t_{vv''} \right] G^R(v''v';\omega) = \delta_{vv'}$$

We need the cases ① $v = v' = l$

② $v \in$ continuum, $v' = l$

① $$\sum_{v'' \in \{l, \text{continuum}\}} \left[\delta_{lv''}(\omega + i\eta) - t_{lv''} \right] G^R(v''l;\omega) = 1$$

$$(\omega + i\eta - \xi_0) G^R(ll;\omega) - \sum_v t_v G^R(vl;\omega) = 1$$

② $$\sum_{v'' \in \{l, \text{contin}\}} \left[\delta_{vv''}(\omega + i\eta) - t_{vv''} \right] G^R(v''l;\omega) = 0$$

$$(\omega + i\eta - \xi_v) G^R(vl;\omega) - t_v^* G^R(ll;\omega) = 0$$

$$\Rightarrow G^R(vl;\omega) = \frac{t_v^* G^R(ll;\omega)}{\omega - \xi_v + i\eta}$$

insert in ①

$$\Rightarrow G^R(ll;\omega) = \frac{1}{\omega - \xi_0 - \sum_v \frac{|t_v|^2}{\omega - \xi_v + i\eta}} = $$

$$= \frac{1}{\omega - \xi_0 - \Sigma^R(\omega)} \quad (*)$$

Although we treat a non-interacting system, we can define a "self-energy" for the single-particle level. It changes the pole structure of the Green function and adds a lifetime to the impurity state.

It comes out of off-diagonal terms in the Hamiltonian
⇒ no longer diagonal in the l-operator
⇒ true diagonal state is superposition of v and l!

(*) compare this to the Green function of an atom hybridized with a fluid conduction bath of electrons (find TRIQS tutorial)!